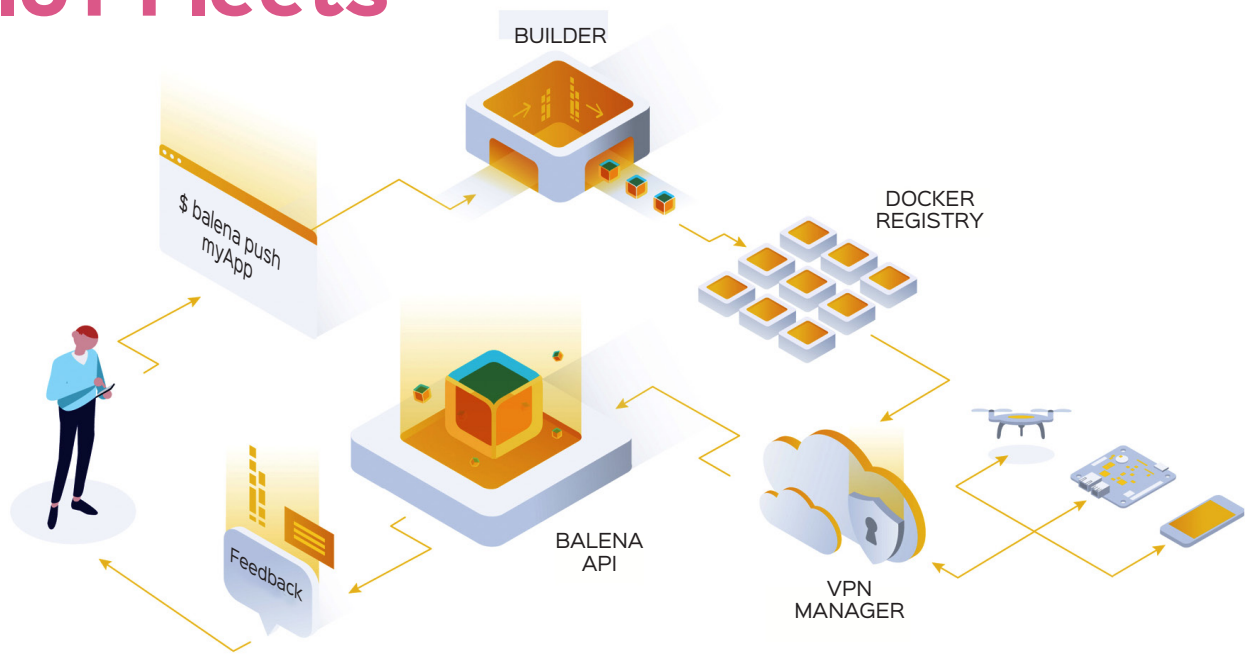


Balena: Helping to Manage IoT Fleets



IoT fleet owners who use numerous connected devices can manage and monitor these with the help of the balena ecosystem. This article outlines how this can be done.

The Internet of Things (IoT) is making the lives of millions of people better every day, whether it is the ease of monitoring your home from the far corners of the world, fitness bands sensing step counts and syncing them with your smartphones, GPS trackers installed inside each truck in a logistics company's fleet giving updates on package movement, or thousands of tiny sensors distributed over a big farm that continuously report soil health for better crop yields.

IoT has become a subtle part of our lives, but like all other good things, there are challenges to keeping the overall experience natural, seamless and secure.

From a manufacturer or IoT fleet owner's point-of-view, there are certain challenges that need to be addressed to ensure the end user experience

is smooth and secure. This includes providing regular software updates on these devices so that all the security vulnerabilities are patched and new features are continuously rolled out.

Many applications involve receiving data from these edge devices deployed in the field, making it extremely important to monitor them remotely for their health status (battery, network, storage), and update them as and when needed. The last thing a fleet owner would want is to send someone to manually update or fix an IoT device which is deployed out in the wild.

Addressing challenges in IoT fleet management

IoT fleet owners need to build a complete infrastructure to remotely monitor and securely update edge devices. They

also need the resources to develop and keep this infrastructure up and running. This can become a pain point when you're managing heterogeneous devices with different architectures (say, x86 and ARM) and different network connectivity types (Ethernet/Wi-Fi), and when you need to add more devices every now and then. As you grow your fleet, you will have thousands of devices and your infrastructure will need to be scaled accordingly.

Setting up this infrastructure in-house takes your focus away from your main application and the problem you are trying to solve as a fleet owner.

What is balena?

Thankfully, there are solutions like balena.io, which is a collection of tools to address the challenges IoT fleet owners